

FESTARA

Collaborative Event Planning and Expense Management

Final Year Project Web Application



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1. Introduction

People have always organized events together. Weddings pull in the whole family. Road trips need everyone to coordinate and split costs. University society events run on committees where a dozen people each handle different things. The problem is not that people do not know how to plan. They have no shared system to do it in.

Most groups manage events across several tools at once. The guest list sits on someone's phone. Expenses get mentioned in a chat message and forgotten by the next day. Tasks are handed out verbally, and two weeks later nobody remembers who was supposed to confirm the venue. When fifteen people are involved and a budget of fifty thousand rupees is on the line, this kind of loose coordination causes real problems. Money goes untracked, responsibilities slip, and the person running everything ends up doing it alone because nobody else has visibility.

Existing software addresses parts of this. Eventbrite [5][9] handles ticketing for public events. Splitwise [10] tracks who owes who after the fact. Professional wedding planning platforms [7] cover everything but charge monthly fees for business use, not for a father organizing his daughter's wedding or five brothers planning a northern areas trip. Research shows that budget management is the single biggest challenge in event planning, with 95% of event planners citing rising costs as their top concern [5]. None of these tools combine event planning, shared expense tracking, and task management in one place that ordinary groups can actually use without a subscription or a steep learning curve [8].

Festara is a web application [1][2] that brings all of this together. A group creates an event, sets a budget, and invites members with specific roles. From that point, everyone works from the same place. Guests get added and tracked. Expenses are logged by whoever spends the money. Tasks are assigned with deadlines. The admin sees everything on one dashboard. The app runs entirely in a web browser on any device, with no installation required and no monthly fee for basic use.

2. Existing System / Related Works

Several tools exist in adjacent spaces. None of them fully address what Festara targets. Studies show that collaboration software with proper access controls and expense tracking is rated as important or highly important by 89% of users [8], yet most event tools either ignore this or charge enterprise prices for it [9]. The table below compares the most relevant existing systems.

App System /	Year	Features	Strength	Limitations
Eventbrite [5][9]	2006	Public event ticketing, registration, attendee tracking	Large reach, polished ticketing infrastructure	No expense tracking, no task management, built for public events not private family coordination
Splitwise [10]	2011	Shared expense tracking, debt splitting, group finance	Clean debt settlement, multi-currency support	No event creation, no guest lists, no task management, free tier severely limited

Aisle Planner [7]	2013	Wedding planning, vendor coordination, budget tracking, guest management	Comprehensive, professional-grade wedding tools	Priced for professional planners, not usable by regular families, steep learning curve
WhatsApp and Google Sheets [6]	N/A	Messaging, manual data entry, file sharing	Free, familiar, no setup needed	No structure, decisions get buried in messages, no role-based control, data easily lost

Table 1: Summary of Existing Systems and Applications

3. Proposed System

Festara is a web application that runs in any browser on desktop or mobile without installation. Groups use it to plan and manage events collaboratively. One person creates the event and takes the admin role. Others join as members or guests, and everyone works from the same shared interface.

The system is built for three types of situations. A family organizing a wedding, where the father acts as admin, his sons as members who contribute to guest lists and expense logs, and relatives as guests who confirm attendance through an invite link. Five brothers on a road trip, where one creates the event and everyone logs what they spend on petrol, accommodation, and food as they go. A university society running an annual event, where the president manages the event, committee members handle assigned tasks, and students register as guests.

The platform does not try to cover everything. It focuses on private, invitation-based events for groups of roughly 3 to 50 people. It is not a public ticketing platform, a payment processor, or a discovery directory. The focus is on the coordination problems that happen inside the group before, during, and after the event.

3.1. Scope

Festara is a web-based application for private, invitation-only events. It covers:

- Family events such as weddings, engagements, Walima, Mehndi, Eid gatherings, and birthday parties
- Friend group activities such as road trips, vacations, and shared outings
- University society events such as tech fests, seminars, annual functions, and society meetings
- Small professional use cases such as independent event planners managing client events without custom software

Outside the scope of this project:

- Public ticketing or paid event registration

- In-app money transfers or payment processing between users
- Virtual event hosting or video streaming
- Enterprise-level CRM, contract management, or marketing tools

3.2. Proposed System Modules

3.2.1. User Authentication Module

- Register with email and password
- Secure login with session management via Supabase Auth [\[1\]](#)
- Password reset through email link
- Profile page to update name and contact details

3.2.2. Event Management Module

- Create an event with name, type, date, location, and description
- Set total event budget at creation
- Edit event details after creation, restricted to Admin
- View all events the logged-in user belongs to from a home dashboard
- Delete event with confirmation prompt, restricted to Admin

3.2.3. Role-Based Access Control Module

- Admin has full control over the event, members, budget, and all settings [\[8\]](#)
- Member can add guests, log expenses, update task status, and view all event data
- Guest receives an invite link, confirms attendance, and views basic event details only
- Admin can change a member's role or remove them from the event
- Every action is restricted based on the assigned role with no manual permission checks required from users

3.2.4. Guest List and RSVP Module

- Add guests individually with name, contact number, and family side such as bride's family or groom's family
- Generate invite links that guests use to confirm their attendance
- Track RSVP status per guest as Pending, Confirmed, or Declined
- Summary view showing total invited, total confirmed, and total pending
- Filter guest list by family side or response status

3.2.5. Expense and Budget Tracking Module

- Total event budget set by Admin and editable throughout planning [\[6\]](#)
- Any member can log an expense with amount, category, description, and date
- Expense categories include Venue, Catering, Decor, Transport, Photography, and Miscellaneous
- Live dashboard showing total budget, total spent, and balance remaining
- Warning indicator when spending crosses 80 percent of budget
- Expense breakdown view by member and by category

3.2.6. Task and Arrangement Module

- Create tasks with title, description, assigned member, and deadline
- Task status tracked through three stages: To Do, In Progress, and Done
- Admin dashboard shows overall task completion as a percentage
- Members see their assigned tasks in a personal view
- All tasks are tied to the event so nothing gets misplaced

3.2.7. Admin Dashboard Module

- Central screen showing event name, date, days remaining, member count, and guest count
- Task completion summary showing done versus pending at a glance
- Budget usage bar showing spent versus remaining
- Recent activity log showing who added or changed what and when
- Quick navigation to all modules from one screen

3.2.8. AI Features Module (Phase 2 - Final Presentation)

- Smart checklist generator: Admin enters event type and guest count, and the AI [\[3\]](#)[\[4\]](#) returns a structured preparation checklist with suggested timelines for each item
- Budget estimator: AI [\[3\]](#) suggests a realistic budget split by category based on event type and expected attendance
- Overspend alert: System flags when a specific expense category is running significantly higher than typical for that event type

3.3. Advantages of the Proposed System

- All event coordination in one place with no switching between WhatsApp, spreadsheets, and phone contacts
- Role-based access prevents confusion about who can do what, Admin has final say, Members contribute, Guests only confirm attendance
- Real-time expense visibility stops budget overruns from going unnoticed
- Task assignments with deadlines replace verbal coordination that routinely falls through
- Free for individual and family use with no subscription barrier
- Works for weddings, trips, and campus events without any reconfiguration
- AI features in Phase 2 give first-time organizers structured guidance without requiring prior experience
- Entirely web-based and works on any device with a browser, no app download required

3.4. Limitations / Disadvantages

- No built-in payment processing: the system tracks debt and spending but does not transfer money between users
- Requires an active internet connection with no offline mode in the initial version
- Guest invites are link-based only with no direct SMS or WhatsApp integration in Phase 1
- Web application only with no dedicated iOS or Android app in the initial release
- AI features in Phase 2 require OpenAI API [\[3\]](#) availability and carry per-request costs
- Data is stored on cloud servers, so users with strict data privacy requirements would need to review the privacy policy

3.5. Application

- Families organizing weddings, engagements, Walima, Mehndi, or large family gatherings
- Friend groups planning road trips, vacations, or group outings where shared costs need tracking
- University societies running annual events, tech fests, orientation programs, and seminars
- Independent event planners and small wedding halls who need a lightweight management tool without the cost of custom software

- Any group of 3 to 50 people coordinating a single event with shared responsibilities and a shared budget

3.6. SDG Goals

Festara maps to the following United Nations Sustainable Development Goals:

- SDG 8 - Decent Work and Economic Growth: Independent event planners and small wedding halls can use the platform to manage their work without purchasing expensive custom software, reducing operational costs for small businesses in the events sector.
- SDG 9 - Industry, Innovation and Infrastructure: The platform applies modern web technologies and AI to a process that most people still manage through informal channels, demonstrating practical software innovation for everyday coordination needs.
- SDG 11 - Sustainable Cities and Communities: Improving how families and communities coordinate shared events contributes to stronger social organization and reduces the financial disputes that poorly managed events often cause.
- SDG 17 - Partnerships for the Goals: The multi-user collaborative model reflects the principle that shared goals work better when roles are clear, responsibilities are visible, and everyone has access to the same information.

4. Hardware / Software Requirements

Software Requirements

- Next.js 16 [\[2\]](#) - React framework for the web frontend with server-side rendering and routing
- Supabase [\[1\]](#) - Backend platform providing PostgreSQL database, user authentication, and real-time data subscriptions
- Vercel [\[2\]](#) - Deployment and hosting platform for the Next.js web application
- OpenAI API [\[3\]](#) - Powers the AI checklist generator and budget estimator in Phase 2
- Vercel AI SDK [\[4\]](#) - Handles streaming AI responses in the web interface
- Tailwind CSS - Utility-first CSS framework for styling the web application
- Node.js 18 and above - JavaScript runtime for the development environment

Hardware Requirements

- Development machine: Any modern laptop or desktop with minimum 8GB RAM and a stable internet connection
- End users: Any device with a modern web browser including desktop, laptop, tablet, or mobile phone

- Server infrastructure: Fully managed by Vercel and Supabase with no dedicated server hardware required

5. References

All references follow IEEE citation format.

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